```
!pip install -q opencv-python matplotlib numpy
```

```
import cv2
import numpy as np
import matplotlib.pyplot as plt
```

```
image_paths = [
    "/content/Picture1.png",
    "/content/Picture2.png",
    "/content/Picture3.png",
    "/content/Picture4.png",
    "/content/Picture5.png"
]

images = []
```

```
for p in image_paths:
    img = cv2.imread(p, 0)
    if img is not None:
        images.append(img)
        print("Loaded:", p)
    else:
        print("Missing:", p)
```

```
Loaded: /content/Picture1.png
Loaded: /content/Picture2.png
Loaded: /content/Picture3.png
Loaded: /content/Picture4.png
Loaded: /content/Picture5.png
```

```
smooth_results = []
sharp_results  = []

for img in images:

    # -------- Smoothing --------
    avg = cv2.blur(img, (5,5))
    gauss = cv2.GaussianBlur(img, (5,5), 1)
    median = cv2.medianBlur(img, 5)
    bilateral = cv2.bilateralFilter(img, 9, 75, 75)

    smooth_results.append([img, avg, gauss, median, bilateral])

    # -------- Sharpening --------
    lap = cv2.Laplacian(img, cv2.CV_64F)
    lap = cv2.convertScaleAbs(lap)
    lap_sharp = cv2.addWeighted(img, 1.0, lap, 1.0, 0)

    blur = cv2.GaussianBlur(img, (9,9), 2)
    unsharp = cv2.addWeighted(img, 1.5, blur, -0.5, 0)

    k = 1.8
    highboost = cv2.addWeighted(img, k, blur, -(k-1), 0)

    sharp_results.append([img, lap_sharp, unsharp, highboost])
```

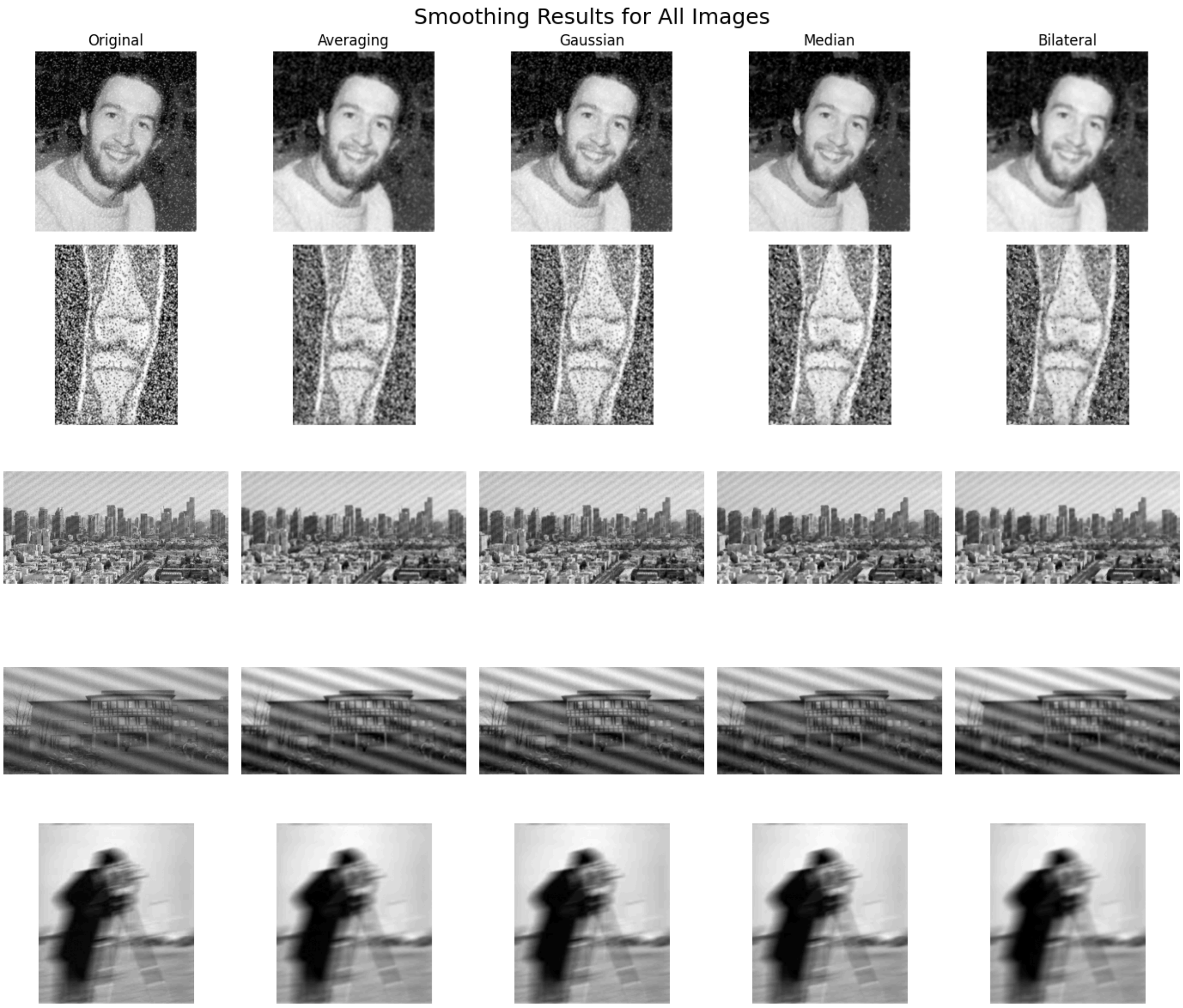
```
titles_smooth = ["Original", "Averaging", "Gaussian", "Median", "Bilateral"]

plt.figure(figsize=(14, 12))
plt.suptitle("Smoothing Results for All Images", fontsize=18)

rows = len(images)
cols = 5

for r in range(rows):
    for c in range(cols):
        plt.subplot(rows, cols, r*cols + c + 1)
        plt.imshow(smooth_results[r][c], cmap='gray')
        if r == 0:
            plt.title(titles_smooth[c], fontsize=12)
        plt.axis("off")

plt.tight_layout()
plt.show()
```



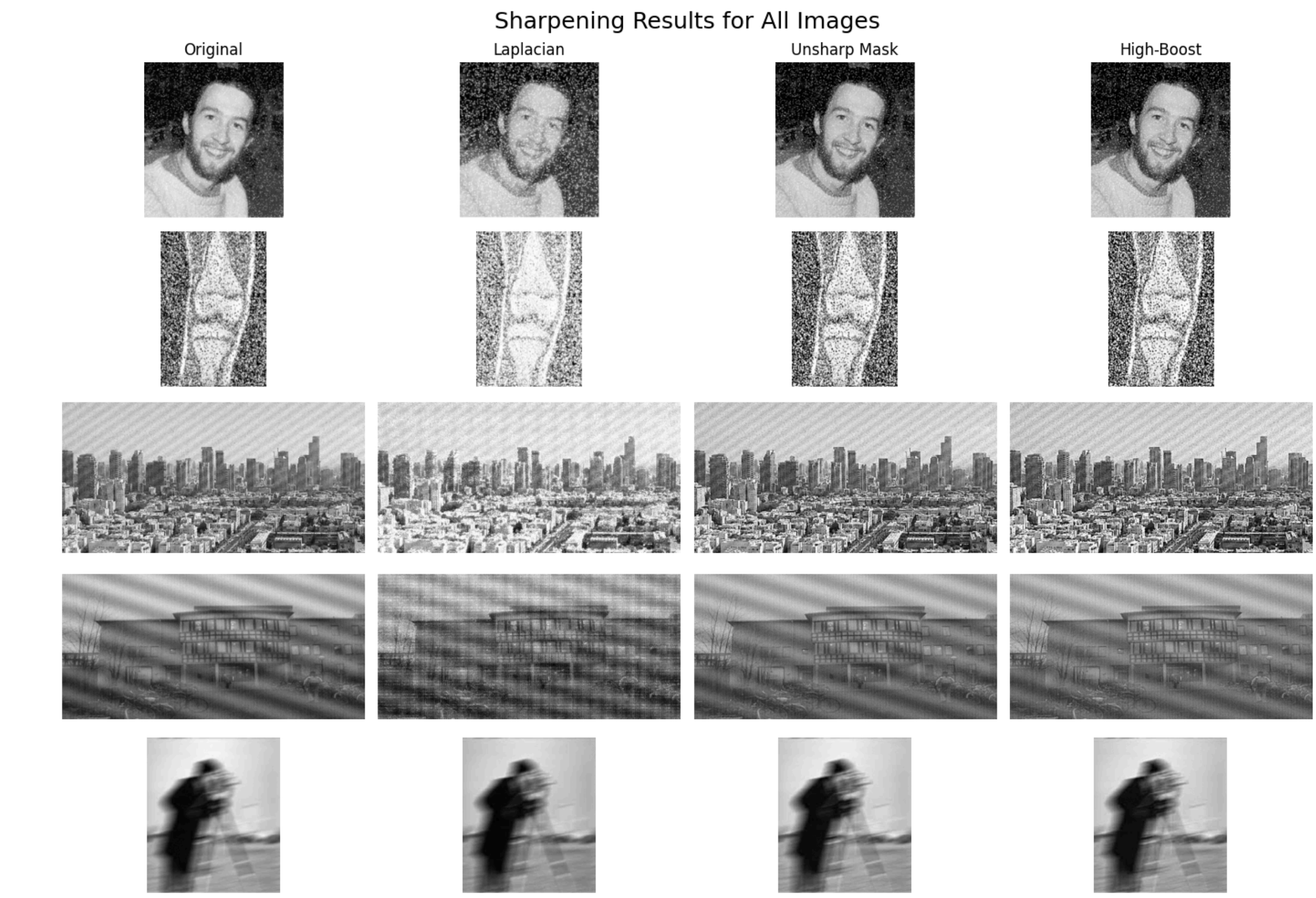

```
titles_sharp = ["Original", "Laplacian", "Unsharp Mask", "High-Boost"]

plt.figure(figsize=(14, 10))
plt.suptitle("Sharpening Results for All Images", fontsize=18)

rows = len(images)
cols = 4

for r in range(rows):
    for c in range(cols):
        plt.subplot(rows, cols, r*cols + c + 1)
        plt.imshow(sharp_results[r][c], cmap='gray')
        if r == 0:
            plt.title(titles_sharp[c], fontsize=12)
        plt.axis("off")

plt.tight_layout()
plt.show()
```


Sharpening Results for All Images
Original
Laplacian
Unsharp Mask
High-Boost